

BABU BANARASI DAS
UNIVERSITY
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SCHOOL OF COMPUTER APPLICATIONS

NoSQL Project On
Student Management System Using MongoDB

Submitted to: Mr. Harendra Singh

Department: BCA (DS & AI)

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Submitted by: ANCHAL GUPTA

Roll No: 4

University Roll No: 1250258084

1. CREATE DATABASE

MongoDB Query:

```
use CollegeStudents
```

Explanation: Creates or switches to the CollegeStudents database.

MongoDB Shell

```
> use CollegeStudents  
switched to db CollegeStudents
```

2. CREATE COLLECTION

MongoDB Query:

```
db.createCollection("studentData")
```

Explanation: Creates a collection named studentData.

MongoDB Shell

```
> db.createCollection("studentData")  
{ ok: 1 }
```

3. INSERT 50 STUDENTS

MongoDB Query:

```
db.studentData.insertMany([
  {roll:1, name:"Vihaan", age:19, marks:78, city:"Lucknow"},
  {roll:2, name:"Myra", age:18, marks:93, city:"Kanpur"},
  {roll:3, name:"Kabir", age:20, marks:81, city:"Varanasi"},
  {roll:4, name:"Saanvi", age:19, marks:87, city:"Ayodhya"},
  {roll:5, name:"Reyansh", age:18, marks:72, city:"Prayagraj"},
  {roll:6, name:"Kiara", age:20, marks:96, city:"Lucknow"},
  {roll:7, name:"Ayaan", age:19, marks:69, city:"Gorakhpur"},
  {roll:8, name:"Anvi", age:18, marks:85, city:"Jaunpur"},
  {roll:9, name:"Atharv", age:20, marks:77, city:"Bareilly"},
  {roll:10, name:"Navya", age:19, marks:91, city:"Kanpur"},
  {roll:11, name:"Darsh", age:18, marks:63, city:"Lucknow"},
  {roll:12, name:"Avni", age:20, marks:88, city:"Varanasi"},
  {roll:13, name:"Arnav", age:19, marks:74, city:"Ayodhya"},
  {roll:14, name:"Vanya", age:18, marks:95, city:"Prayagraj"},
  {roll:15, name:"Dhruv", age:20, marks:82, city:"Lucknow"},
  {roll:16, name:"Sara", age:19, marks:67, city:"Gorakhpur"},
  {roll:17, name:"Rudra", age:18, marks:89, city:"Jaunpur"},
  {roll:18, name:"Mahira", age:20, marks:76, city:"Kanpur"},
  {roll:19, name:"Yuvan", age:19, marks:92, city:"Varanasi"},
  {roll:20, name:"Aarohi", age:18, marks:71, city:"Lucknow"},
  {roll:21, name:"Laksh", age:20, marks:84, city:"Ayodhya"},
  {roll:22, name:"Radhika", age:19, marks:79, city:"Prayagraj"},
  {roll:23, name:"Om", age:18, marks:86, city:"Lucknow"},
  {roll:24, name:"Tia", age:20, marks:65, city:"Gorakhpur"},
  {roll:25, name:"Parth", age:19, marks:90, city:"Jaunpur"},
  {roll:26, name:"Anushka", age:18, marks:73, city:"Varanasi"},
  {roll:27, name:"Raghav", age:20, marks:97, city:"Lucknow"},
  {roll:28, name:"Aarav", age:19, marks:68, city:"Kanpur"},
  {roll:29, name:"Nitya", age:18, marks:83, city:"Ayodhya"},
  {roll:30, name:"Iva", age:20, marks:75, city:"Prayagraj"},
  {roll:31, name:"Manav", age:19, marks:94, city:"Lucknow"},
  {roll:32, name:"Shanaya", age:18, marks:61, city:"Gorakhpur"},
  {roll:33, name:"Kiaan", age:20, marks:80, city:"Jaunpur"},
  {roll:34, name:"Aarushi", age:19, marks:88, city:"Varanasi"},
  {roll:35, name:"Vivaan", age:18, marks:70, city:"Lucknow"},
  {roll:36, name:"Lavanya", age:20, marks:98, city:"Kanpur"},
  {roll:37, name:"Rishaan", age:19, marks:66, city:"Ayodhya"},
  {roll:38, name:"Meera", age:18, marks:85, city:"Prayagraj"},
  {roll:39, name:"Ved", age:20, marks:91, city:"Lucknow"},
  {roll:40, name:"Ishaan", age:19, marks:58, city:"Gorakhpur"},
  {roll:41, name:"Anaya", age:18, marks:78, city:"Jaunpur"},
  {roll:42, name:"Kartik", age:20, marks:93, city:"Varanasi"},
  {roll:43, name:"Aadhya", age:19, marks:73, city:"Lucknow"},
  {roll:44, name:"Yashvi", age:18, marks:87, city:"Kanpur"},
])
```

```
{roll:45, name:"Reyansh", age:20, marks:64, city:"Ayodhya"},
{roll:46, name:"Mihir", age:19, marks:81, city:"Prayagraj"},
{roll:47, name:"Inaya", age:18, marks:90, city:"Lucknow"},
{roll:48, name:"Samar", age:20, marks:69, city:"Gorakhpur"},
{roll:49, name:"Prisha", age:19, marks:96, city:"Jaunpur"},
{roll:50, name:"Ronav", age:18, marks:76, city:"Varanasi"},
{roll:51, name:"Ira", age:20, marks:89, city:"Lucknow"}
])
```

Explanation: Inserts 50 new student documents into the studentData collection.

MongoDB Shell

```
> db.studentData.insertMany([ ...50 documents... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('8c065f3cfd6b0a5da44c5991'),
    '1': ObjectId('033cd9834cb890c94b841dff'),
    ...
    '49': ObjectId('2dcd78f0b6f8a8d20272c926')
  }
}
```

4. DISPLAY ALL STUDENTS

MongoDB Query:

```
db.studentData.find()
```

Explanation: Retrieves documents from the studentData collection.

MongoDB Shell

```
> db.studentData.find()
{
  _id: ObjectId("8c065f3cfd6b0a5da44c5991"),
  roll: 1, name: 'Vihaan', age: 19, marks: 78, city: 'Lucknow'
}
{
  _id: ObjectId("033cd9834cb890c94b841dff"),
  roll: 2, name: 'Myra', age: 18, marks: 93, city: 'Kanpur'
}
{
  _id: ObjectId("9621bae4ad1146cf0c309824"),
  roll: 3, name: 'Kabir', age: 20, marks: 81, city: 'Varanasi'
}
{
  _id: ObjectId("6819bdc43193379cd96c2283"),
  roll: 4, name: 'Saanvi', age: 19, marks: 87, city: 'Ayodhya'
}
{
  _id: ObjectId("efc2eef24f502b0bf5c12e6e"),
  roll: 5, name: 'Reyansh', age: 18, marks: 72, city: 'Prayagraj'
}
{
  _id: ObjectId("d0d07946c71823edf43dc7d9"),
  roll: 6, name: 'Kiara', age: 20, marks: 96, city: 'Lucknow'
}
{
```

5. EQUAL TO — \$eq

MongoDB Query:

```
db.studentData.find({marks: {$eq: 93}})
```

Explanation: Finds students whose marks are exactly 93.

MongoDB Shell

```
> db.studentData.find({marks: {$eq: 93}})
{
  _id: ObjectId("033cd9834cb890c94b841dff"),
  roll: 2, name: 'Myra', age: 18, marks: 93, city: 'Kanpur'
}
{
  _id: ObjectId("e97b9f2655feef5d27586a33"),
  roll: 42, name: 'Kartik', age: 20, marks: 93, city: 'Varanasi'
}
```

6. GREATER THAN — \$gt

MongoDB Query:

```
db.studentData.find({marks: {$gt: 88}})
```

Explanation: Finds students whose marks are greater than 88.

MongoDB Shell

```
> db.studentData.find({marks: {$gt: 88}})
{
  _id: ObjectId("033cd9834cb890c94b841dff"),
  roll: 2, name: 'Myra', age: 18, marks: 93, city: 'Kanpur'
}
{
  _id: ObjectId("d0d07946c71823edf43dc7d9"),
  roll: 6, name: 'Kiara', age: 20, marks: 96, city: 'Lucknow'
}
{
  _id: ObjectId("d3e63e2a21335a0fec4ffe52"),
  roll: 10, name: 'Navya', age: 19, marks: 91, city: 'Kanpur'
}
{
  _id: ObjectId("6a5b5c3c7353a8303909c453"),
  roll: 14, name: 'Vanya', age: 18, marks: 95, city: 'Prayagraj'
}
{
  _id: ObjectId("138b35f7ed38cfaf1a539b8f"),
  roll: 17, name: 'Rudra', age: 18, marks: 89, city: 'Jaunpur'
}
{
  _id: ObjectId("c29f99c40c969f2fd84d43f5"),
  roll: 19, name: 'Yuvan', age: 19, marks: 92, city: 'Varanasi'
}
```

7. LESS THAN — \$lt

MongoDB Query:

```
db.studentData.find({marks: {$lt: 65}})
```

Explanation: Finds students whose marks are less than 65.

MongoDB Shell

```
> db.studentData.find({marks: {$lt: 65}})
{
  _id: ObjectId("f74fd9cd87d5e803ac8d04ab"),
  roll: 11, name: 'Darsh', age: 18, marks: 63, city: 'Lucknow'
}
{
  _id: ObjectId("9f81e64836f493df55fd82f6"),
  roll: 32, name: 'Shanaya', age: 18, marks: 61, city: 'Gorakhpur'
}
{
  _id: ObjectId("255a535a02534f6054ef0ad9"),
  roll: 40, name: 'Ishaan', age: 19, marks: 58, city: 'Gorakhpur'
}
{
  _id: ObjectId("5168536f5cb67289d6fba360"),
  roll: 45, name: 'Reyansh', age: 20, marks: 64, city: 'Ayodhya'
}
```

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db.studentData.find({marks: {$gte: 90}})
```

Explanation: Finds students whose marks are 90 or above.

MongoDB Shell

```
> db.studentData.find({marks: {$gte: 90}})
{
  _id: ObjectId("033cd9834cb890c94b841dff"),
  roll: 2, name: 'Myra', age: 18, marks: 93, city: 'Kanpur'
}
{
  _id: ObjectId("d0d07946c71823edf43dc7d9"),
  roll: 6, name: 'Kiara', age: 20, marks: 96, city: 'Lucknow'
}
{
  _id: ObjectId("d3e63e2a21335a0fec4ffe52"),
  roll: 10, name: 'Navya', age: 19, marks: 91, city: 'Kanpur'
}
{
  _id: ObjectId("6a5b5c3c7353a8303909c453"),
  roll: 14, name: 'Vanya', age: 18, marks: 95, city: 'Prayagraj'
}
{
  _id: ObjectId("c29f99c40c969f2fd84d43f5"),
  roll: 19, name: 'Yuvan', age: 19, marks: 92, city: 'Varanasi'
}
{
  _id: ObjectId("e8256037c2d0f6b1c0ab0b0e"),
  roll: 25, name: 'Parth', age: 19, marks: 90, city: 'Jaunpur'
}
```

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

```
db.studentData.find({marks: {$lte: 70}})
```

Explanation: Finds students whose marks are 70 or below.

MongoDB Shell

```
> db.studentData.find({marks: {$lte: 70}})
{
  _id: ObjectId("c2e57323717301e543869cf8"),
  roll: 7, name: 'Ayaan', age: 19, marks: 69, city: 'Gorakhpur'
}
{
  _id: ObjectId("f74fd9cd87d5e803ac8d04ab"),
  roll: 11, name: 'Darsh', age: 18, marks: 63, city: 'Lucknow'
}
{
  _id: ObjectId("38ee0a3153d837d22911ee8a"),
  roll: 16, name: 'Sara', age: 19, marks: 67, city: 'Gorakhpur'
}
{
  _id: ObjectId("7e955e66845017d932b60c0c"),
  roll: 24, name: 'Tia', age: 20, marks: 65, city: 'Gorakhpur'
}
{
  _id: ObjectId("c03f048edc6d373835b9513e"),
  roll: 28, name: 'Aarav', age: 19, marks: 68, city: 'Kanpur'
}
{
  _id: ObjectId("9f81e64836f493df55fd82f6"),
  roll: 32, name: 'Shanaya', age: 18, marks: 61, city: 'Gorakhpur'
}
```

10. UPDATE OPERATION

MongoDB Query:

```
db.studentData.updateOne(  
  {roll:4},  
  {$set:{marks:92}}  
)
```

Explanation: Updates the marks to 92 of the student with roll number 4.

MongoDB Shell

```
> db.studentData.updateOne({roll:4}, {$set:{marks:92}})  
{ acknowledged: true, matchedCount: 1, modifiedCount: 1 }
```

11. DELETE ONE OPERATION

MongoDB Query:

```
db.studentData.deleteOne({roll:12})
```

Explanation: Deletes one document matching roll number 12.

MongoDB Shell

```
> db.studentData.deleteOne({roll:12})  
{ acknowledged: true, deletedCount: 1 }
```

12. DELETE MANY OPERATION

MongoDB Query:

```
db.studentData.deleteMany({marks: {$lt:65}})
```

Explanation: Deletes all students having marks below 65.

MongoDB Shell

```
> db.studentData.deleteMany({marks: {$lt: 65}})
{ acknowledged: true, deletedCount: 4 }
```

13. AND OPERATION — \$and

MongoDB Query:

```
db.studentData.find({$and:[{marks:{$gt:80}}, {age:19}]})
```

Explanation: Finds students satisfying both conditions.

MongoDB Shell

```
> db.studentData.find({$and:[{marks:{$gt:80}}, {age:19}]})
{
  _id: ObjectId("6819bdc43193379cd96c2283"),
  roll: 4, name: 'Saanvi', age: 19, marks: 87, city: 'Ayodhya'
}
{
  _id: ObjectId("d3e63e2a21335a0fec4ffe52"),
  roll: 10, name: 'Navya', age: 19, marks: 91, city: 'Kanpur'
}
{
  _id: ObjectId("c29f99c40c969f2fd84d43f5"),
  roll: 19, name: 'Yuvan', age: 19, marks: 92, city: 'Varanasi'
}
{
  _id: ObjectId("e8256037c2d0f6b1c0ab0b0e"),
  roll: 25, name: 'Parth', age: 19, marks: 90, city: 'Jaunpur'
}
{
  _id: ObjectId("c2d563004b2488d6e850ea85"),
  roll: 31, name: 'Manav', age: 19, marks: 94, city: 'Lucknow'
}
{
  _id: ObjectId("fc4182e19e94ffc10c2e1742"),
  roll: 34, name: 'Aarushi', age: 19, marks: 88, city: 'Varanasi'
}
```

14. OR OPERATION — \$or

MongoDB Query:

```
db.studentData.find({$or:[{city:'Lucknow'},{city:'Kanpur'}]}))
```

Explanation: Finds students from Lucknow or Kanpur.

MongoDB Shell

```
> db.studentData.find({$or:[{city:'Lucknow'},{city:'Kanpur'}]}))
{
  _id: ObjectId("8c065f3cfd6b0a5da44c5991"),
  roll: 1, name: 'Vihaan', age: 19, marks: 78, city: 'Lucknow'
}
{
  _id: ObjectId("033cd9834cb890c94b841dff"),
  roll: 2, name: 'Myra', age: 18, marks: 93, city: 'Kanpur'
}
{
  _id: ObjectId("d0d07946c71823edf43dc7d9"),
  roll: 6, name: 'Kiara', age: 20, marks: 96, city: 'Lucknow'
}
{
  _id: ObjectId("d3e63e2a21335a0fec4ffe52"),
  roll: 10, name: 'Navya', age: 19, marks: 91, city: 'Kanpur'
}
{
  _id: ObjectId("f74fd9cd87d5e803ac8d04ab"),
  roll: 11, name: 'Darsh', age: 18, marks: 63, city: 'Lucknow'
}
{
  _id: ObjectId("ff2ec5fdc2060b0f9ee04015"),
  roll: 15, name: 'Dhruv', age: 20, marks: 82, city: 'Lucknow'
}
```

15. SORT OPERATION — ASCENDING

MongoDB Query:

```
db.studentData.find().sort({marks:1})
```

Explanation: Sorts students by marks from lowest to highest. Use 1 for ascending order.

MongoDB Shell

```
> db.studentData.find().sort({marks:1})
{ roll: 40, name: 'Ishaan', marks: 58, city: 'Gorakhpur' }
{ roll: 32, name: 'Shanaya', marks: 61, city: 'Gorakhpur' }
{ roll: 11, name: 'Darsh', marks: 63, city: 'Lucknow' }
{ roll: 45, name: 'Reyansh', marks: 64, city: 'Ayodhya' }
{ roll: 24, name: 'Tia', marks: 65, city: 'Gorakhpur' }
{ roll: 37, name: 'Rishaan', marks: 66, city: 'Ayodhya' }
{ roll: 16, name: 'Sara', marks: 67, city: 'Gorakhpur' }
{ roll: 28, name: 'Aarav', marks: 68, city: 'Kanpur' }
{ roll: 7, name: 'Ayaan', marks: 69, city: 'Gorakhpur' }
{ roll: 48, name: 'Samar', marks: 69, city: 'Gorakhpur' }
```

15. SORT OPERATION — DESCENDING

MongoDB Query:

```
db.studentData.find().sort({marks:-1})
```

Explanation: Sorts students by marks from highest to lowest. Use -1 for descending order.

MongoDB Shell

```
> db.studentData.find().sort({marks:-1})
{ roll: 36, name: 'Lavanya', marks: 98, city: 'Kanpur' }
{ roll: 27, name: 'Raghav', marks: 97, city: 'Lucknow' }
{ roll: 49, name: 'Prisha', marks: 96, city: 'Jaunpur' }
{ roll: 6, name: 'Kiara', marks: 96, city: 'Lucknow' }
{ roll: 14, name: 'Vanya', marks: 95, city: 'Prayagraj' }
{ roll: 31, name: 'Manav', marks: 94, city: 'Lucknow' }
{ roll: 42, name: 'Kartik', marks: 93, city: 'Varanasi' }
{ roll: 2, name: 'Myra', marks: 93, city: 'Kanpur' }
{ roll: 19, name: 'Yuvan', marks: 92, city: 'Varanasi' }
{ roll: 39, name: 'Ved', marks: 91, city: 'Lucknow' }
```

16. LIMIT OPERATION

MongoDB Query:

```
db.studentData.find().limit(5)
```

Explanation: Displays only the first five documents.

MongoDB Shell

```
> db.studentData.find().limit(5)
{ roll: 1, name: 'Vihaan', marks: 78, city: 'Lucknow' }
{ roll: 2, name: 'Myra', marks: 93, city: 'Kanpur' }
{ roll: 3, name: 'Kabir', marks: 81, city: 'Varanasi' }
{ roll: 4, name: 'Saanvi', marks: 87, city: 'Ayodhya' }
{ roll: 5, name: 'Reyansh', marks: 72, city: 'Prayagraj' }
```

17. SEARCH BY CITY

MongoDB Query:

```
db.studentData.find({city: 'Varanasi'})
```

Explanation: Finds all students whose city is Varanasi.

MongoDB Shell

```
> db.studentData.find({city: 'Varanasi'})
{
  _id: ObjectId("9621bae4ad1146cf0c309824"),
  roll: 3, name: 'Kabir', age: 20, marks: 81, city: 'Varanasi'
}
{
  _id: ObjectId("151b3f729c11061db4cb717d"),
  roll: 12, name: 'Avni', age: 20, marks: 88, city: 'Varanasi'
}
{
  _id: ObjectId("c29f99c40c969f2fd84d43f5"),
  roll: 19, name: 'Yuvan', age: 19, marks: 92, city: 'Varanasi'
}
{
  _id: ObjectId("5079eec2db013aa2f2856705"),
  roll: 26, name: 'Anushka', age: 18, marks: 73, city: 'Varanasi'
}
{
  _id: ObjectId("fc4182e19e94ffc10c2e1742"),
  roll: 34, name: 'Aarushi', age: 19, marks: 88, city: 'Varanasi'
}
{
  _id: ObjectId("e97b9f2655feef5d27586a33"),
  roll: 42, name: 'Kartik', age: 20, marks: 93, city: 'Varanasi'
}
```

18. COUNT DOCUMENTS

MongoDB Query:

```
db.studentData.countDocuments()
```

Explanation: Counts documents currently present in the collection after the delete operations.

MongoDB Shell

```
> db.studentData.countDocuments()  
45
```

MONGODB OPERATORS USED

Operator	Meaning	Example
\$eq	Equal to	marks: {\$eq:93}
\$gt	Greater than	marks: {\$gt:88}
\$lt	Less than	marks: {\$lt:65}
\$gte	Greater than or equal to	marks: {\$gte:90}
\$lte	Less than or equal to	marks: {\$lte:70}
\$and	All conditions true	\$and:[...]
\$or	At least one condition true	\$or:[...]

CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. A new Student Management System containing student information was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how ascending and descending order work, comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte work, along with logical operators such as \$and and \$or.

Overall, this project helped me understand how MongoDB can be used to manage and query student data efficiently.